Do Now · retrieval

Lesson 124 · Procedural Programming: Modules and State

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which paradigm describes a desired result, rather than detailed step-by-step changes?

Q2

A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

Q3

For the binary value 1010, what is its decimal value?

Q4

A linked-list node has data and what other essential field?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Sequence and control — Statements execute in an order; selection and loops direct flow.

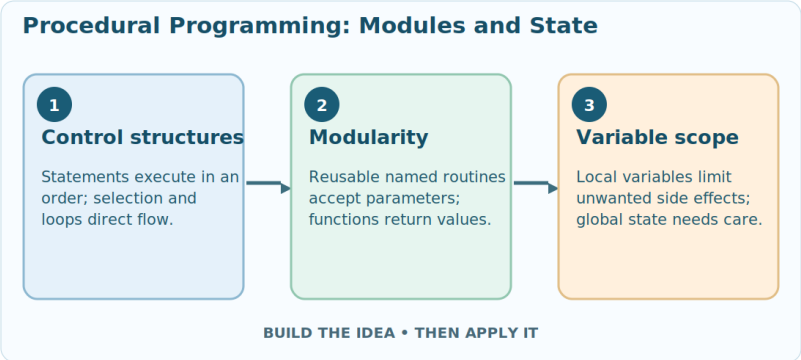
Core idea 2

Procedure/function — Reusable named routines accept parameters; functions return values.

Core idea 3

Scope and state — Local variables limit unwanted side effects; global state needs care.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

procedure — Named subroutine mainly performing an action.

function — Subroutine that returns a value.

parameter — Named input used by a routine.

scope — Region where an identifier may be accessed.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

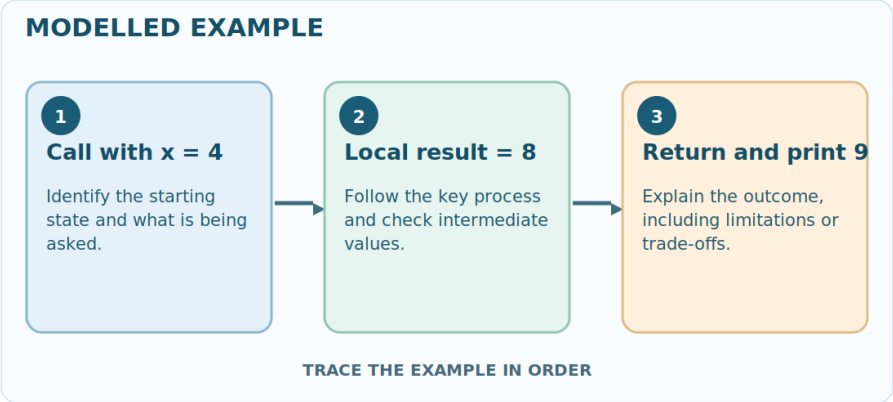
Problem

Trace FUNCTION doublePlusOne(x): result=x*2; RETURN result+1; print(doublePlusOne(4)).

Follow the worked method

1. Model answer — Parameter x=4, local result=8, return value 9, printed output 9. The local result does not become a global variable merely because the function runs. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain the benefits of using functions/procedures rather than one long main routine. [6]

Question 2

Trace FUNCTION f(n): a=n+2; RETURN a*3; OUTPUT f(2). [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain local/global scope and give a bug caused by modifying shared global state. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A local variable declared inside a function is automatically accessible everywhere in the program.

Use key vocabulary from memory

1. Explain procedure in your own words.

2. Explain function in your own words.

3. Explain parameter in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.